

Parametric modeling of Heat Sink - FreeCAD Asset 01

1. Requirements

To develop the parametric model of heat sink using FreeCAD software. A heat sink is a passive cooling device that absorbs and disperses thermal energy away from a hot electronic component into the surrounding air. The parametric model consists of an aluminum heat sink of dimension 18mm x 18mm x 2mm which is called heat sink base. The heat sink is mounted directly on top of the CPU/package. Vertical rectangular fins of 2mm x 2mm x 2mm are modeled on the top of the heat sink base. 16 fins are placed equidistant from each other.

CAD requirements

- **Heat sink base:** 18 mm × 18 mm × 2 mm rectangular base.
- **Cooling fins:** Repeated vertical cooling fins arranged uniformly on the heat sink base.
- **Material:** Aluminum.
- **Base dimensions:** The overall 18 mm × 18 mm base dimensions must remain unchanged.
- **Connectivity:** All fins must remain physically connected to the heat sink base.
- **Parametric design:** Fin dimensions including fin height, thickness and spacing must be editable through FreeCAD parameters or features.
- **Fin count:** Total 16 fins are placed on the top of the heat sink placed at 2mm from heat sink base boundary
- **Valid geometry:** The final heat sink must consist of a single connected solid with no unintended gaps or disconnected bodies.

2. Required Engineering Work

Starting from the provided FreeCAD project, complete the heat-sink thermal simulation.

1. Create the heat-sink geometry

- Develop a parametric heat-sink model in FreeCAD.
- Use an 18 × 18 × 2 mm aluminum base.
- Include 16 uniformly distributed vertical cooling fins.

2. Inspect the geometry

- Check the heat-sink geometry for modeling errors.
- Verify that the base and fins form a valid, connected solid.
- Ensure there are no unintended gaps, overlaps, or disconnected bodies.

3. Define parametric dimensions

- Base length: 18 mm
- Base width: 18 mm

- Base thickness: 2 mm
- Fin thickness: 2 mm
- Fin height: 2 mm
- Fin spacing: 2 mm
- Number of fins: 16

4. Establish parametric relationships

- Ensure the sketches show 0 degrees of freedom / Fully constrained in FreeCAD.
- Define the dimensions as editable parameters.
- Ensure that changing a parameter automatically updates the associated geometry.
- Use appropriate FreeCAD features, constraints, expressions, or a Spreadsheet to control the parameters.

5. Verify the fin arrangement

- Ensure all 16 fins remain uniformly distributed.
- Ensure every fin remains physically connected to the base after parameter changes.
- Maintain the required 18×18 mm base dimensions.

3. Acceptance Requirements

- The FreeCAD project opens without errors.
- The heat-sink base is $18 \times 18 \times 2$ mm.
- The model contains 16 vertical cooling fins.
- Fin thickness, fin height, spacing, and number of fins are defined parametrically.
- The relevant sketches are fully constrained (0 degrees of freedom).
- Changing the designated parameter, such as fin height, automatically updates the heat-sink geometry.
- The 16 fins remain uniformly distributed and physically connected to the base after parameter changes.
- The final heat sink is a valid connected solid with no unintended gaps or disconnected bodies.
- The aluminum material is correctly assigned.
- The model can be recomputed successfully after the parametric modification.
- The final model matches the golden solution requirements and can be saved as a valid *.FCStd file.

4. Deliverable

Parametric FreeCAD model with a volume of 776 mm^3